ALPER KALLE

Ph.D. Student in Computer Science

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Research Interests: Efficient Deep Learning · LLM Compression · Tensor Decomposition and Quantization · Edge AI

Education

2024 – Present **Ph.D. in Computer Science**, *CEA (French Alternative Energies and Atomic Energy Commission) - Paris-Saclay University*
Thesis: *Tensor Methods for Deep Neural Network Compression*

2021 – 2023 **Master in Applied Mathematics**, *Toulouse III - Paul Sabatier University*

2017 – 2021 **Bachelor in Mathematics**, *Galatasaray University*

Publications

2026 M.R.M. Boudjemaa, **A. Kalle**, X. Mai, J.H.M. Goulart, C. Févotte. "Whitening Spherical Gaussian Mixtures in the Large-Dimensional Regime". *IEEE International Conference on Acoustics, Speech, and Signal Processing* (Oral Presentation). **[Paper]**

2025 **A. Kalle**, T. Rudkiewicz, M.O. Ouerfelli, M. Tamaazousti. "Distribution-Aware Tensor Decomposition for Compression of Convolutional Neural Networks". *Advances in Neural Information Processing Systems (NeurIPS)*. **[Code][Paper]**

Research & Teaching Experience

Mar – Sep 2023 **Research Intern (Master)**, *IRIT & IMT*, Toulouse
Topic: *Analysis of tensor decomposition methods for machine learning in the large-data regime.*
- Advisors: Henrique Goulart (IRIT) and Xiaoyi Mai (IMT).

2024/2025 & 2025/2026 **Teaching Assistant**, *Polytech Paris-Saclay*, Paris
Teaching assistant for the Mathematics 2 course.

Summer Schools & Training

Jun 2025 **A Journey through Deep Learning**, *MaLGa Center, University of Genoa*, Italy
40-hour intensive PhD-level school covering fundamentals of deep learning, CNNs, transformers, and generative models.

Skills

Deep Learning PyTorch, TensorFlow, CUDA/cuDNN, Model Optimization
Hardware GPU Programming, ONNX, TensorRT, Edge Deployment
Programming Python, C#, MATLAB, SQL, Git, Slurm
Languages Turkish (Native), English (Fluent), French (Fluent)

Selected Talks

Oct 2025 **Seminar**, *CEA*, Paris
Title: *Compression of Neural Networks by Tensor Decomposition*

Sept 2025 **Speech**, *Junior Conference on Data Sciences and Engineering*, Paris
Title: *Data-Driven Tensor Decomposition: A New Approach to Compressing CNNs*

Jun 2023 **Seminar**, *IRIT*, Toulouse
Title: *Analysis of Tensor-Based Gaussian Mixture Model Estimation*

Awards & Scholarships

2021 – 2022 **Scholarship of Master Galatasaray**, *Embassy of France in Turkey*

2018 – 2021 **TUBITAK-2205 Excellence Scholarship**, *TUBITAK (Turkey)*
Merit scholarship for top undergraduate students in exact sciences.